

Project

Additive Bachelier model – Calibration and Monte Carlo

Financial Engineering

You are asked to calibrate two different models from the European options on WTI oil options prices on the 2nd of June 2020 and simulate the processes.

The folder “Data” contains

- The folders “datacalls” and “dataputs”, that contain the files with the mid prices of call (and put) options for the different maturity (the filename corresponds to the option’s expiry, each file contains prices on all the grid of strikes for each value date).
 - The file “Expiries_Futures”, with the expiry date of each option.
- 1) Calibrate the Forward and Discount Factors following the technique described in [2]. Hereinafter, based on these calibrated values, consider zero-rates as a deterministic function of time.
 - 2) Calibrate the Additive Bachelier model (characteristic function in (2) of [3]), with $\alpha = \frac{1}{3}$, with initial guess for the parameters $\eta_0 = 0.2$ and $k_0 = 1$. Follow the procedure described in section 3 of [3].
 - 3) Calibrate the Additive Logistic model (for real valued underlying, described in [4]).
 - 4) Consider the following exotic pay-off

$$1_{\left\{K > \frac{1}{N} \sum_{i=1}^N F_{t_i, t_N}\right\}}$$

where t_i $i = 1, \dots, N$ are the expiries of the quoted European options (with t_N equal to the 16th November 2022) and F_{t_i, t_N} is defined as the forward expiring in t_N valued at time t_i . The strike K is equal to F_{t_0, t_N} . Price the exotic option with the two models. The notional is 20 Million \$.

Hint: Simulate the paths using the Lewis-FFT algorithm (the steps are described in Appendix B of [1]). As described in [1], the CDF should be estimated using FFT (similarly to the one when pricing a Digital option).

- 5) [Facultative] Hedge the product with plain vanilla options and the underlying front future (establish a quantitative rule). The hedging cost (bid-ask) is 1bp for the underlying and 6bp for each option. Check the quality of the hedge on the next two Tuesdays: control the total hedging cost and the P&L.
- 6) [Facultative] The hedge can be repeated on the next four Tuesdays (using the same rule). It will be provided four additional dates of the test set on the 5th of June 2025 to check the quality of the hedge as in point 5. On the 7th of June at 18:00, send a synthetic document with only the results.

Hint:

Once you have estimated the CDF and simulated the random vector of uniform U , we suggest using MATLAB `interp1` to perform the numerical inversion of the CDF (i.e. `interp1` implements all the algorithm steps in [1, Appendix B] after the simulation of U). Notice that MATLAB's `interp1` has the option 'spline'. Pay attention when "extrapolating" outside the range with the `interp1` function (some suggestions are available in the paper).

References

- [1] Azzone, M., & Baviera, R. (2023). A fast Monte Carlo scheme for additive processes and option pricing. *Computational Management Science*, 20(1), 31.
- [2] Azzone, M., & Baviera, R. (2021). Synthetic forwards and cost of funding in the equity derivative market. *Finance Research Letters*, 41, 101841.
- [3] Baviera, R., & Massaria, M.D. (2025). The Additive Bachelier model with an application to the oil option market in the Covid period
- [4] Carr, P., & Torricelli, L. (2021). Additive logistic processes in option pricing. *Finance and Stochastics*, 25(4), 689-724.

Realize a library in Matlab. Optional Python.

Room: Massaria

Delivery address: financial.engineering.polimi@gmail.com.